# LUKE THOMPSON

UNREAL ENGINE 5 GAMEPLAY PROGRAMMER (C++)

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin · GitHub: github.com/XylarDark

## PROFESSIONAL SUMMARY

Gameplay programmer building HomeWorld solo in Unreal Engine 5.8 and C++ since February 2026: character movement, Gameplay Ability System abilities, interaction, NPC behaviour, and designer-tunable data-driven systems, backed by C++ automation tests. Shipped Shift, a playable Phaser 3 / TypeScript game, and did earlier contract game work in C# at OutFusion. Self-directed remote worker in Canada (US tax-treaty country), Eastern Time.

## CORE SKILLS

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|---|---|
| **Languages** | C++, C#, Python, TypeScript/JavaScript, SQL |
| **Unreal Engine 5** | Gameplay Ability System, custom CharacterMovementComponent modes, Enhanced Input, Blueprints, AI controllers, PCG, Automation tests |
| **Other engines** | Phaser 3, Unity |
| **Workflow** | Git/GitHub pull requests, Visual Studio, Python editor tooling, playtesting and bug verification |

## SELECTED PROJECTS

**HomeWorld — Unreal Engine 5.8 / C++ game** — Feb 2026 – present

Independent (solo) · UE 5.8, C++, Python · private repository, available on request

- Built gameplay in C++ for a third-person ARPG survival and family-sim: character, Enhanced Input, a time-of-day subsystem that gates body and spirit forms, inventory, crafting, and six GAS abilities (interact, place, heal, spirit shield and burst, protector attack) on a C++ base class made for Blueprint children.
- Wrote a custom glide movement mode by subclassing the single CharacterMovementComponent (`PhysCustom`: forward drive against a fixed sink rate, time-constant steering), plus a traversal component with sprint, trace-based mantle and vault, spirit blink with cooldown, and a fall soft-reset that probes for landing ground. Fixed a same-tick re-entry stack overflow in the glide-to-walk landing hand-off.
- Made systems designer-editable without code changes: tuning exposed as editor properties, interactions as a priority-ordered verb table (a new prop is a row, not a branch), and seeded, deterministic zone generation read from JSON zone specs. Added NPC follow and escort AI, a beast-taming state flow, and a spirit-stealth light and alert system.
- Kept stability up with 42 C++ automation test files on a shared scoped game-world fixture (no leaked worlds between tests), a versioned save system that migrates old saves and refuses newer ones, and Python editor and playtest tooling.

**Shift — playable shop & delivery game** — Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame · Repo: github.com/XylarDark/ShiftGame

- Shipped a complete browser/PWA game loop with a fixed 60 Hz simulation and render interpolation, cached A* routing with auto-drive, and a tiered render budget that keeps it smooth on phones.

**Portfolio Studio** — Sep 2026 – present

Independent · React, TypeScript, Supabase · xylardark.github.io/Portfolio-Studio

- Shipped a public media-résumé site with an editor, embeddable work cards, and per-employer pages.

## RELEVANT EXPERIENCE

**Video Game Developer · Contract · OutFusion · Hamilton, ON** — 2022

- Integrated and tested game UI art assets across mobile and desktop; validated UX through iterative playtesting.
- Built and tested C# REST API integrations for player data; partnered on bug verification and release quality.

**Software Developer · 12-Month Contract · TruckSail · Ancaster, ON** — 2020

- Implemented STM32/CAN firmware with defect finding for reliability; built JS/Bootstrap UIs and optimized SQL; maintained GitHub workflows.

## ADDITIONAL EXPERIENCE

**Delivery Driver (Uber; Kindling Cannabis) · Hamilton, ON** — 2022 – 2026

- Independent schedule while building a game-development portfolio in 2026 (HomeWorld, Shift, Portfolio Studio).

## EDUCATION

**Diploma, Computer Systems Technology — Software Development** · Mohawk College · Hamilton, ON · 2019

Remote · Eastern Time (UTC−5 / UTC−4) · Canada